

Chang Liu

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EDUCATION

Ph.D. in Computer & Information Science & Engineering	Syracuse University, 2027 (Exp.)
M.S. in Computer Science	Syracuse University, 2023
B.S. in Computer Science	Syracuse University, 2021

TECHNICAL SKILLS

Programming Languages: Python, Rust, C, x86 Assembly, Rocq
Frameworks & Tools: CompCert, LLVM, Ghidra, Docker, PyTorch, AWS, SQLite

RESEARCH INTERESTS

Foundations: Logic programming, Static analysis, Compilers, Machine learning
Applications: Reverse engineering, Vulnerability detection, LLM-based binary analysis

PUBLICATIONS

*: equal contributions

- [1] **Chang Liu***, Rebecca Saul*, Yihao Sun, Edward Raff, Maya Fuchs, Townsend Southard Pantano, James Holt, Kristopher Micinski. "Assemblage: Automatic Binary Dataset Construction for Machine Learning." (**NeurIPS D&B**, 2024)
- [2] Rebecca Saul, **Chang Liu**, Noah Fleischmann, Richard J Zak, Kristopher Micinski, Edward Raff, James Holt. "Is Function Similarity Over-Engineered? Building a Benchmark." (**NeurIPS D&B**, 2024)
- [3] **Chang Liu**, Noah Fleischmann, Nicolò Altamura, Edward Raff, James Holt, Kristopher Micinski. "Assemblage-DeepHistory: A Cross-Build Binary Dataset with Temporal Coverage." (Preprint, 2026)
- [4] **Chang Liu**, Yihao Sun, Thomas Gilray, Kristopher Micinski. "Superset Decompilation." (Preprint, 2026)

EXPERIENCES

- Research Assistant** Syracuse University, 2023 - Present
- Led the design, development, and maintenance of a large-scale machine executable dataset of 1.7M binaries and 10TB data, used by research communities and OpenAI
 - Designed and implemented a modular declarative decompiler, lifting Linux x86-64 ELF binaries to C through a pipeline of CompCert IRs (35K LoC Rust and Datalog)
 - Built an LLM-agent benchmark for binary vulnerability identification on 329 CVEs, exposing cross-build generalization gaps in frontier LLMs
- Teaching Assistant** Syracuse University, 2020 - 2022
- Served as teaching assistant and grader for 4 computer science courses (CIS 352 Programming Languages: Theory and Practice, CIS 487 Access Control, Security and Trust)
 - Held weekly office hours and helped students with programming assignments
- Automated Program Grading Web Service** autograde.org, 2020 - 2022
- Developed and maintained a Docker-based web platform with Vue.js, MariaDB, and Nginx
 - Platform adopted in 10+ curricula and served 1000+ students

SERVICES AND AWARDS

Reviewing: NeurIPS, AAAI AICS Workshop
Awards: Merit Scholarship (50% Tuition Coverage), Syracuse University, 2021
Mentoring: Alex Duly (M.S., Syracuse University)